MLFF-DATA9A7e Cross-System Qualification

mdstats project

2026-07-30

1. Purpose

MLFF-DATA9A7e qualifies the generalization completed in DATA9A7a–DATA9A7d. The stage does not add a new physical descriptor or declare that a fitted model is scientifically valid. It proves that the same DATA4–DATA7 preparation path can be exercised for materially different profile classes while preserving profile lineage, generic feature ownership, optional-extension isolation, and selection evidence.

The required bounded cases are:

1. a generic crystalline solid;
2. an amorphous solid;
3. a homogeneous liquid;
4. a multiphase interface;
5. a crystalline material with the optional LTA extension.

The first four cases must neither import the MLFF LTA implementation modules nor serialize legacy LTA top-level fields. The fifth case must activate the explicit `porous_network -> zeolite -> lta` chain and carry LTA results only through the generic optional-extension envelopes.

2. Scope and non-goals

DATA9A7e owns software-path qualification. It verifies:

- explicit material-profile contracts;
- DATA4–DATA7 digest lineage;
- phase/geometry plan realization;
- universal structural-feature and event activation;
- optional-extension activation or absence;
- nonempty deterministic DATA7 selection evidence;
- generic import isolation;
- canonical serialization without legacy LTA fields;
- immutable qualification records and suite evidence.

DATA9A7e does **not** establish:

- physical realism of the synthetic bounded fixtures;
- transferability of a trained potential;
- convergence of RDF, coordination, VDOS, diffusion, or other observables;
- production corpus sufficiency;
- final feature weights, acceptance thresholds, or checkpoint choice;
- performance or scaling for large systems.

Those responsibilities remain with the owning analysis manuals, DATA9A8 observable-comparison policy, and later DATA9B execution.

3. Qualification records

3.1 Clean-import evidence

`ImportIsolationEvidence` binds a clean-interpreter probe:

- probe identity;
- whether the interpreter started clean;
- material-specific modules present before generic import;
- material-specific modules present after generic MLFF import;
- forbidden module prefixes;
- digest of the probe script;
- a derived pass/fail result.

For DATA9A7e the forbidden generic prefixes are:

```
mdstats.training_data.lta_profile
mdstats.training_data.lta_selection
```

Importing `mdstats`, constructing generic profiles, and using generic DATA4–DATA7 APIs must not load either module. Accessing a public LTA symbol may load the requested module lazily.

3.2 Per-case qualification evidence

`CrossSystemQualificationCaseRecord` binds:

- case ID and case kind;
- qualification-policy digest;
- material-profile and contract digests;
- declared phase kinds and geometry;
- exact DATA4, DATA5, DATA6, and DATA7 bundle digests;
- phase/geometry plan digest;
- enabled universal feature and event families;
- active optional-extension IDs;
- selected-frame count;
- clean-import evidence digest;
- forbidden imported modules;
- forbidden serialized field paths;
- profile, lineage, and selection gates;
- final derived pass/fail state.

The record cannot be marked passed unless every required component passes.

3.3 Suite evidence

`CrossSystemQualificationSuiteRecord` requires one record for every policy- required case kind. All cases must share the same policy digest. The suite pass state is the conjunction of its case pass states. JSON write/read operations are canonical and digest-verified.

4. Case-specific requirements

4.1 Generic crystalline solid

Required:

- exactly one `crystalline_solid` phase;
- a realized phase/geometry plan;
- universal structural features in DATA6;
- nonempty DATA7 selection;
- no `lta` extension;

- no forbidden MLFF LTA imports;
- no legacy LTA serialization fields.

4.2 Amorphous solid

Required:

- exactly one `amorphous_solid` phase;
- disordered-phase radial and local-density defaults;
- universal structural features and generic events;
- the same generic isolation requirements as the crystal case.

4.3 Liquid

Required:

- exactly one `liquid` phase;
- liquid phase defaults and generic structural selection;
- no assumption of rings, sites, or cations;
- the same generic isolation requirements.

4.4 Multiphase interface

Required:

- `interface` geometry;
- at least two declared phases;
- explicit phase and interface atom groups;
- phase-composed feature and event defaults;
- geometry-aware group priorities;
- no implicit surface or interface inference;
- the same generic isolation requirements.

4.5 LTA extension

Required:

- a crystalline phase;
- explicit `porous_network`, `zeolite`, and `lta` extensions;
- canonical `lta` partition and selection catalogs inside generic `ProfileFeatureCatalog` envelopes;
- no legacy LTA top-level fields in new serialization;
- nonempty DATA7 selection combining universal and extension feature blocks.

5. Import architecture

The generic package import path must not import MLFF LTA implementations merely because the public API exposes optional LTA symbols. DATA9A7e therefore changes LTA exports to lazy resolution:

```
import mdstats
    -> generic training-data modules loaded
    -> lta_profile and lta_selection absent

mdstats.LtaSelectionPolicy
    -> lta_selection loaded on demand
```

`data4_bundle.py` and `data6_bundle.py` use type-checking imports plus local runtime imports only when legacy LTA payloads are decoded or an active LTA policy is requested.

6. Bounded end-to-end workflow

Each qualification case executes:

```
explicit profile contracts
        -> DATA4 raw/profile evidence
        -> DATA5 partition and leakage audit
        -> DATA6 profile-aware structural evidence
        -> DATA7 fitted metric and deterministic selection
        -> qualification case record
```

The fixture may be small, but the same public builders and lineage checks used by production workflows must be exercised. Test-only bypasses of bundle validation are forbidden.

7. Serialization audit

Generic cases recursively scan canonical DATA4, DATA6, and DATA7 payloads for legacy keys:

```
lta_partition_features
lta_selection_features
```

The current canonical schemas must not emit these fields. Generic cases must also contain no active extension whose ID is `lta`. Historical schema readers remain supported and are tested separately; compatibility does not permit new canonical evidence to regress to legacy fields.

8. Failure semantics

Qualification fails closed for:

- missing explicit material-profile contracts;
- missing phase/geometry plan;
- DATA4–DATA7 digest mismatch;
- wrong phase or geometry for the declared case;
- LTA extension in a generic case;
- missing LTA hierarchy or catalog in the LTA case;
- missing DATA7 selection;
- absent clean-interpreter import evidence;
- forbidden generic imports;
- legacy LTA fields in new generic serialization;
- incomplete suite case coverage;
- tampered qualification records.

9. Regression requirements

The focused release gate must include:

- all five DATA4–DATA7 bounded workflows;
- clean top-level import without MLFF LTA modules;
- lazy LTA-symbol loading;
- generic rejection of forbidden import evidence;
- universal-plus-extension DATA7 metric fitting;
- suite completeness and round-trip serialization;
- DATA4/DATA6 historical compatibility tests;
- DATA9A7a–DATA9A7d regression tests;
- dependency-graph acyclicity and documentation assertions;
- source/wheel public API parity.

10. Acceptance criteria

DATA9A7e is complete when:

1. all five cases pass the immutable suite policy;
2. generic cases import and serialize no MLFF LTA implementation evidence;
3. the LTA case uses only generic extension envelopes in canonical bundles;
4. DATA4–DATA7 lineage and nonempty selection are proven for every case;
5. the clean source tree and installed wheel expose equivalent qualification APIs;
6. the MLFF architecture and dependency graph identify DATA9A7e as implemented;
7. no physical-observable algorithm has moved into the MLFF branch.

11. Next stage

DATA9A8 introduces profile-aware observable comparison policies. It consumes analysis-owned physical results through the standardized observable bridge, freezes comparison rules before evaluation, aggregates by condition and atom group, and supports checkpoint decisions without reimplementing the analyses.